How-To Library

Informatica

Deploying Big Data Management 10.2.1 on the AWS Cloud Platform through the Amazon Marketplace

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Abstract

Customers of Amazon Web Services (AWS) and Informatica can deploy Informatica Big Data Management® 10.2.1 through the AWS marketplace. The automated marketplace solution fully integrates Big Data Management with the AWS platform and the Amazon EMR cluster. The installed solution includes several pre-configured mappings that you can use to discover the capabilities of Big Data Management to load, transform, and write data to various AWS storage resources.

Supported Versions

- Big Data Management 10.2.1

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Overview

Customers of Amazon Web Services and Informatica can deploy a big data solution that fully integrates Big Data Management with the AWS cloud platform and the Amazon EMR cluster.

Several different methods are available for deploying Big Data Management:

Hybrid deployment

Install and configure the Informatica domain and Big Data Management on-premise, and configure them to push processing to the Amazon EMR cluster.

Manual cloud deployment

Manually install and configure the Informatica domain and Big Data Management on AWS EC2 instances in the same region as your Amazon EMR cluster, or deploy the domain on-premises.

Marketplace cloud deployment

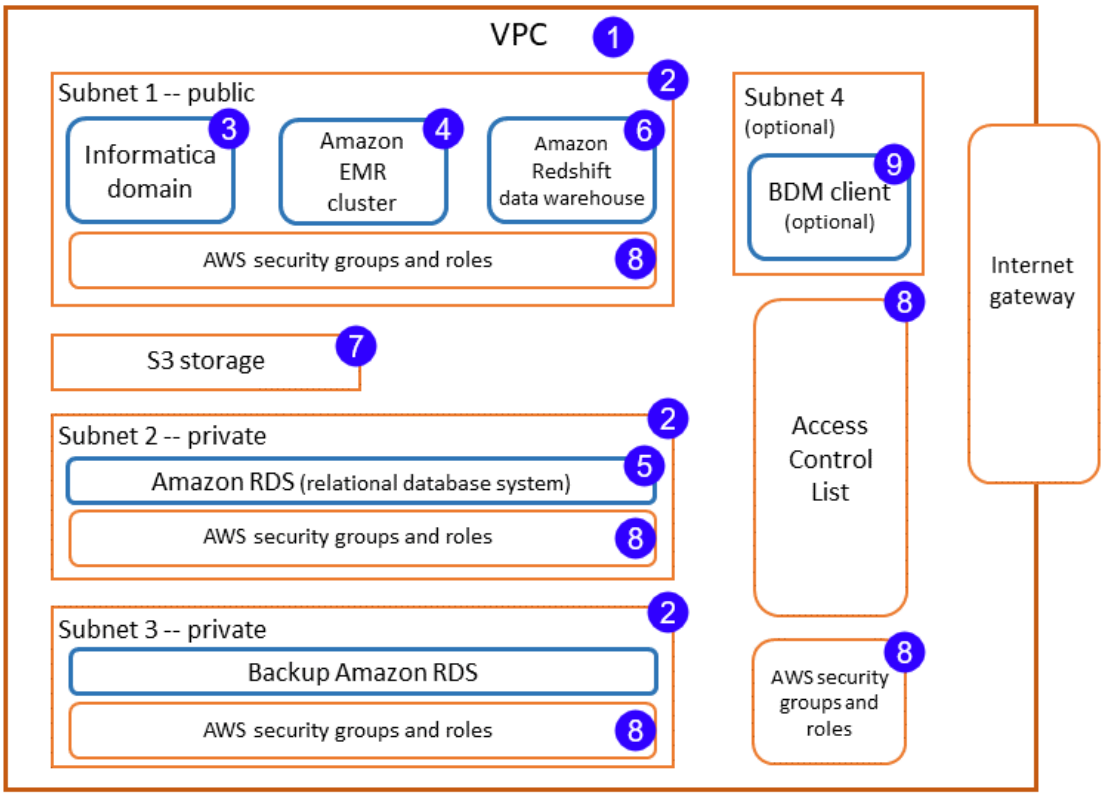
Execute a Big Data Management deployment from the AWS marketplace to create an Informatica domain and an Amazon EMR cluster in the AWS cloud, exploring Big Data Management functionality through prepackaged mappings.

The Big Data Management marketplace solution on AWS enables you to automate the deployment of the Informatica domain, an Amazon EMR cluster, and integration with storage and other resources in the AWS cloud platform. the marketplace solution includes prepackaged mappings that demonstrate various Big Data Management functionality.

The architecture of the marketplace solution on the AWS platform is essentially the same as the manual implementation of a Big Data Management cloud deployment as shown in the image in the previous section.

When you configure the implementation of the Amazon marketplace Big Data Management solution, you make choices about the size of EC2 instances on the AWS platform.

The following diagram shows the architecture of the Big Data Management on AWS marketplace solution:



The numbers in the architecture diagram correspond to items in the following list:

1. A virtual public cloud (VPC) to contain the Big Data Management deployment.
2. Subnets to contain specific elements of the deployment. Create one public subnet and at least one private subnet, each in a different availability zone.
3. The Informatica domain, including the Model Repository Service and the Data Integration Service.
4. Amazon EMR cluster to process mappings and other jobs from the Data Integration Service.

5. Amazon RDS databases for Informatica domain repositories:
   - Domain repository database
   - Model repository
   - Monitoring Model repository
6. An Amazon Redshift data warehouse, to act as a repository for data sources and targets.
7. S3 storage, to act as a temporary location for files that the data integration service moves between EC2 instances and the EMR cluster.
8. Access control lists to contain authentication and authorization settings.
9. Additional storage space on the cluster.
10. Big Data Management clients, such as the developer tool or an infacmd console. See "Informatica clients" on page 5 for an explanation of each of these.

The Big Data Management Solution on the AWS Marketplace

The solution includes fully configured AWS resources including S3 storage, Amazon RDS databases for repositories, an Amazon EMR cluster for processing, and an Informatica domain populated with sample data and mappings. Optionally, you can also include an Amazon Redshift data warehouse and a remote Windows server with Informatica clients.

Informatica Domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

Application Services

Model Repository Service

The Model Repository Service manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata. The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication.

The Informatica domain can run several other services. For more information about Informatica services, see the *Informatica Application Service Guide*.

Domain Repositories

Informatica repositories, posted on SQL databases, store metadata about domain objects. Informatica repositories include the following:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows.

Monitoring Model repository

The monitoring Model repository stores statistics for Data Integration Service jobs. You configure the monitoring Model Repository Service in the domain properties.

In addition to these domain repositories, the solution also requires a repository for Hive metadata. This repository is hosted on an SQL database. It stores Hive table metadata to enable Hadoop operations.

For more information about domain repositories, see the *Informatica Application Service Guide*.

Informatica clients

You can use several different clients with Informatica Big Data Management:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

AWS Platform Elements

When you deploy the Big Data Management solution on the AWS cloud platform, the resulting environment consist of many elements. Some of these elements must pre-exist the implementation process, and others are created by the automated implementation.

This section describes all of the possible elements that the final environment can contain.

AWS Building Blocks

Amazon Web Services offers the basic building blocks of storage, networking, and computation, as well as services such as a managed database, big data, and messaging services.

A Big Data Management deployment on EMR can use the following service offerings:

Amazon EC2 instances

Amazon Elastic Compute Cloud (Amazon EC2) instances provide scalable computing capacity in the Amazon Web Services (AWS) cloud. You can launch as many or as few virtual servers as you need with zero investment on hardware. You can configure security and networking, and manage storage. Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic.

Big Data Management can be deployed on Amazon EC2 with the ability to scale up and scale down the environment based on requirements. Big Data Management can be deployed in a mixed environment that contains on-premises machines and Amazon EC2 instances.

Amazon S3 storage

Amazon Simple Storage Service (S3) is easy-to-use object storage with a simple web service interface to store and retrieve any amount of data from anywhere on the web.

Big Data Management provides native, high-volume connectivity to Amazon S3 and support for Hive on S3. It is designed and optimized for big data integration between cloud and on-premise data sources to S3 as object stores.

Amazon Redshift

Amazon Redshift is a cloud-based, fast, fully managed, petabyte-scale data warehouse that makes it simple and cost-effective to analyze all data using existing business intelligence tools. Informatica's PowerExchange for Amazon Redshift connector allow users to securely read data from or write data to Amazon Redshift.

Amazon RDS

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks, freeing you up to focus on your applications and business. Amazon RDS offers several database engines to choose from; the Big Data Management deployment uses the SQL Server database.

Amazon Virtual Private Cloud

Amazon Virtual Private Cloud (Amazon VPC) lets you provision a logically isolated section of the Amazon Web Services (AWS) cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including selection of your own IP address range, creation of subnets, and configuration of route tables and network gateways.

You can easily customize the network configuration for your Amazon Virtual Private Cloud. For example, you can create a public-facing subnet for your web servers that has access to the Internet, and place your back end systems such as databases or application servers in a private facing subnet with no Internet access. You can leverage multiple layers of security, including security groups and network access control lists, to help control access to Amazon EC2 & EMR instances in each subnet.

AWS Regions and Availability Zones

Regions are self-contained geographical locations where AWS services are deployed. Regions have their own deployment of each service. Each service within a region has its own endpoint that you can interact with to use the service.

Regions contain availability zones, which are isolated fault domains within a general geographical location. Some regions have more availability zones than others. While provisioning, you can choose specific availability zones or let AWS select them for you.

Networking, Connectivity, and Security

Amazon AWS enables the following networking, connectivity, and security features:

Virtual Private Cloud (VPC)

VPC has several different configuration options. See the VPC documentation for a detailed explanation of the options and choose based on your networking requirements. You can deploy Big Data Management in either public or private subnets.

Connectivity to the Internet and Other AWS Services

Deploying the instances in a public subnet allows them to have access to the Internet for outgoing traffic as well as to other AWS services, such as S3 and RDS.

Private Data Center Connectivity

You can establish connectivity between your data center and the VPC hosting your Informatica services by using a VPN or Direct Connect. We recommend using Direct Connect so that there is a dedicated link between the two networks with lower latency, higher bandwidth, and enhanced security. You can also connect to EC2 through the Internet via VPN tunnel.

Security Groups

You can define rules for EC2 instances and define allowable traffic, IP addresses, and port ranges. Instances can belong to multiple security groups.

AWS Direct Connect

AWS Direct Connect makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect, you can establish private connectivity between AWS and your datacenter, office, or co-location environment, which in many cases can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than Internet-based connections.

Using AWS Direct Connect with the Big Data Management deployment is optional.

Understanding Amazon Cluster Types and Big Data Management Deployment Options

The central component of Amazon EMR is the cluster. A cluster is a collection of Amazon Elastic Compute Cloud (Amazon EC2) instances. Each instance in the cluster is called a node. Each node has a role within the cluster, referred to as the node type. Amazon EMR also installs layers of software components on each node type, giving each node a role in a distributed application like Apache Hadoop.

Amazon Node Types

Amazon EMR nodes are of the following types:

Master node

Manages the cluster by running software components which coordinate the distribution of data and tasks among other nodes—so-called slave nodes—for processing. The master node tracks the status of tasks and monitors the health of the cluster.

Core node

A slave node that has software components which run tasks and store data in the Hadoop Distributed File System (HDFS) on the cluster.

Task node

An optional slave node that has software components which only run tasks.

Amazon EMR Layers

Amazon EMR service architecture consists of several layers, each of which provides certain capabilities and functionality to the cluster. An EMR cluster has the following layers:

Storage

The storage layer includes the different file systems that are used with your cluster. You can choose from among the following storage options:

- Hadoop Distributed File System (HDFS) is a distributed, scalable file system for Hadoop. HDFS distributes the data it stores across instances in the cluster, storing multiple copies of data on different instances to ensure that no data is lost if an individual instance fails. This ephemeral storage that is reclaimed when you terminate a cluster. HDFS is useful for caching intermediate results during MapReduce processing or for workloads which have significant random I/O.
- EMR File System (EMRFS). Amazon EMR uses the EMR File System to enable Hadoop to access data stored in Amazon S3 as if it were a file system like HDFS. You can use either HDFS or Amazon S3 as the file system in your cluster. Most often, Amazon S3 is used to store input and output data, and intermediate results are stored in HDFS.
- Local File System. The local file system refers to a locally connected disk. When you create a Hadoop cluster, each node is created from an Amazon EC2 instance that comes with a pre-configured block of pre-attached disk storage called an instance store. Data on instance store volumes persists only during the life cycle of its Amazon EC2 instance.

Cluster resource management

The resource management layer is responsible for managing cluster resources and scheduling the jobs for processing data.

By default, Amazon EMR uses YARN (Yet Another Resource Negotiator), which is a component introduced in Apache Hadoop 2.0 to centrally manage cluster resources for multiple data-processing frameworks. However, there are other frameworks and applications that are offered in Amazon EMR that do not use YARN as a resource manager. Amazon EMR also has an agent on each node which administers YARN components, keeps the cluster healthy, and communicates with the Amazon EMR service.

Data processing frameworks

The data processing framework layer is the engine used to process and analyze data. Many frameworks run on YARN or have their own resource management.

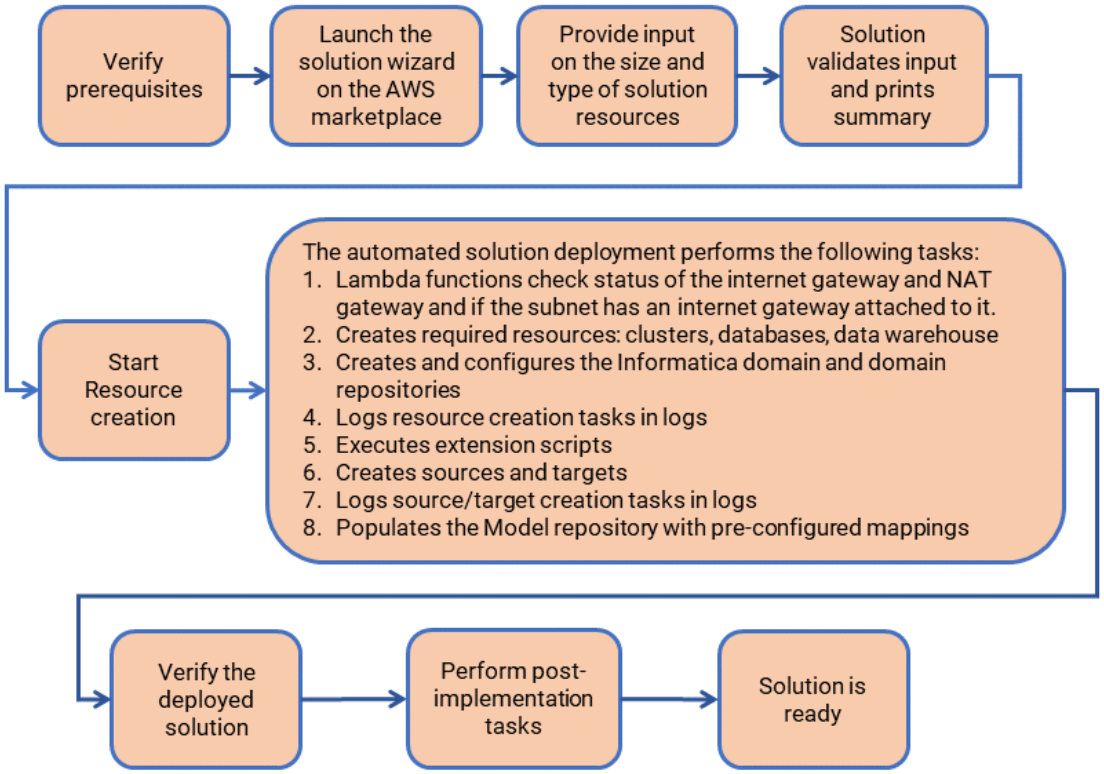
Informatica documentation refers to these data processing frameworks as run-time engines. The engine you choose depends on processing needs, such as batch, interactive, in-memory, or streaming. Your choice of run-time engine affects the languages and interfaces on the application layer, which is the layer used to interact with the data you want to process.

The following main run-time engines are available for Amazon EMR:

- **Spark.** Apache Spark is a cluster framework and programming model for processing big data workloads. Like Hadoop MapReduce, Spark is an open-source, distributed processing system but uses directed acyclic graphs for execution plans and leverages in-memory caching for datasets. Spark supports multiple interactive query modules such as SparkSQL
- **Blaze.** Informatica Blaze is the industry's unique data processing engine integrated with YARN to provide intelligent data pipelining, job partitioning, job recovery, and scalability, which is optimized to deliver high performance, scalable data processing leveraging Informatica's cluster aware data integration technology.

Implementation Overview

The following diagram shows how the marketplace solution is implemented:



Pre-Implementation Tasks

Before you start the automated Big Data Management deployment, perform the following pre-implementation tasks:

- Verify prerequisites.
- Gather information about the AWS account and environment.

Verify Prerequisites

Verify the following prerequisites before you begin to provision the AWS marketplace solution:

- Big Data Management license. The marketplace offering is a "bring your own license" (BYOL) solution. Get the license from the email message from Informatica and upload it to S3 storage on your AWS account.
- AWS architecture. Create a Virtual Private Cloud (VPC) with three subnets: one public subnet, and two private subnets, each in different availability zones.
- AWS key pair. The solution uses the key pair for authentication and authorization
- Access to the AWS console and permissions to create the resources that the Big Data Management solution requires.

Gather AWS Account and Environment Information

Before you begin to provision the Big Data Management marketplace solution stack on AWS, gather the following information about the AWS account and environment:

Property	Description
AWS_ACCESS_KEY	AWS access key for the user account to use to initialize the stack.
AWS_SECRET_ACCESS_KEY	AWS secret access key for the user account to initialize the stack.
REGION	AWS region in which to create the stack.
EMR_ROLE	AWS EMR full access role. This role must be assigned to the user account that initializes the stack before this script is executed.
EC2_INSTANCE_PROFILE	AWS ec2 access policy. This policy must be assigned to the user account that initializes the stack.
EC2_KEY_PAIR	2048-bit SSH-2 RSA key pair to use for stack security. For more information about key pair requirements and how to generate the key pair, see Amazon documentation.
EC2_SUBNET	AWS subnet id. This subnet should be present in the VPC in which the stack is created.
MASTER_SECURITY_GROUP	AWS security group for master EMR node. The user account that initializes the stack must be assigned to this security group.
SLAVE_SECURITY_GROUP	AWS security group for slave EMR node. The user account that initializes the stack must be assigned to this security group.
HADOOP_NODE_JDK_HOME	AWS EMR JDK path. Required to modify the Hadoop connection for mappings run in Blaze mode.

Provision the Big Data Management on AWS Marketplace Solution

Use the AWS Marketplace website to provision AWS platform resources including a Big Data Management deployment.

Launch the Implementation Wizard

Log on to the AWS marketplace site and select **Cloud Formation** > **Stacks** > **Create Stack** to begin the process of deploying the solution.

Populate the properties on the **Create Stack** page. The following sections describe the properties to populate.

Specify Details section

In this section, specify a stack name for the solution on the AWS platform.

Parameters section

In this section, specify network parameters for the solution.

The following table describes values for parameters in the **Network Configuration** subsection:

Parameter	Description
VPC ID	ID of the Virtual Private Cloud (VPC) in which to deploy the solution. The VPC must meet the following requirements: - Set up with public access through the internet via an attached internet gateway. - The DNS Resolution property of the VPC must be set to Yes. - The Edit DNS Hostnames property of the VPC must be set to Yes.
Informatica domain subnet	ID of a publically available subnet in which to deploy the Informatica domain.
Informatica database subnets	Specify the IDs of three different subnets. One of these subnets must meet the following requirements: - Must be a member of the VPC. - Must have access to the internet via gateway. The Informatica domain must use this subnet. - Must have auto-assign public IPv4 enabled. The other two subnets must meet the following requirements: - Must be members of the VPC. - Must be in two separate regions, to enable database failover. - May be private or public. Any public subnet must have auto-assign public IPv4 enabled. It is not necessary to choose subnets in the domain subnet.
IP address range	IP address range to use to limit SSH access from the Informatica domain to the EC2 instance. The address range is rendered as a Classless Inter-Domain Routing (CIDR) address block. For example, to specify the range of 10.20.30.0 to 10.20.30.255, enter the following string: `10.20.30.40/24`

The following table describes values for parameters in the **Amazon EC2 Configuration** subsection:

Parameter	Description
Key pair name	Select an existing EC2 KeyPair name to enable SSH access for Informatica services to the EC2 instance.

Informatica Big Data Management Configuration section

In this section, specify parameters for the Big Data Management domain. The following table describes values for parameters in this section:

Parameter	Description
Informatica domain instance type	EC2 instance type for the Informatica domain to reside on. Default is m4.large. **Note:** When you select an instance type here and in later steps, be aware that Amazon charges more when you select a larger instance type.
Informatica domain name	Type a name for the Informatica domain.
Informatica node name	Type a node name for Informatica domain instances. This string will be appended as a prefix to domain instances.

Parameter	Description
Informatica administrator user name	Type a name for the Informatica domain administrator. In this field and the following field, you can specify any user name and password. Make a note of the user name and password, and use it later to log in to the Administrator tool to configure the Informatica domain.
Informatica administrator password	Type a password for the Informatica domain administrator. Retype this password in the next field.
Encryption key phrase	Text string to serve as a base word from which to generate an encryption key for the Informatica domain.
BDM license key location	Enter the location of the Big Data Management license key file. The location is the name of the S3 bucket where the key was saved. **Note:** Do not include the license file name in the path. For example: `myBucketName`
BDM license key name	Enter the path and filename of the Big Data Management license key file in the S3 bucket location. The path must include subdirectories *under* the bucket name. For example, where the entire path including the bucket name is `myBucketName/SubDir1/SubDir2/BDMLicense.key`, type the following: `SubDir1/SubDir2/BDMLicense.key`

Amazon RDS configuration for Informatica domain database section

In this section, specify parameters for the Amazon RDS relational database to host the Informatica domain. The following table describes values for parameters in this section:

Parameter	Description
Database type	Amazon RDS database type. Default is SQLServer-Standard-Edition-2016-License. Choose the default database type.
Database name for Informatica domain	Type a name for the Informatica domain repository database.
Database name for Model Repository Service	Type a name for the Model repository database.
Database username	Type a user name for the domain and the Model repository database. In this field and the following field, you can specify any user name and password.
Database password	Type a password for the domain repository database user. Retype this password in the next field.

Amazon EMR configuration

In this section, specify parameters for the Amazon EMR cluster. The following table describes values for parameters in this section:

Parameter	Description
EMR cluster name	Type a name for the EMR cluster.
EMR master instance type	Select the instance type. Each type corresponds to a different size, in ascending order of size. Default is m4.xlarge.
EMR core instance type	Choose an instance type for EMR cluster core nodes. Default is m4.xlarge.
EMR core nodes	Enter an integer for the number of core nodes to support the Big Data Management deployment. Minimum is 1. Maximum is 500. **Note:** Informatica requires at least a minimum of 8 CPU vCores and 30 GB memory.

Amazon Redshift configuration

In this section, specify parameters for a cluster for the Amazon Redshift data warehouse.

Note: Configuring a data warehouse for the solution is optional. if you choose No for the first value, ignore the rest of the values in this section.

The following table describes values for parameters in this section:

Parameter	Description
Redshift cluster needed	Whether to create an Amazon Redshift data warehouse or not. Default is No. If you choose Yes, populate values for the remaining parameters in this section.
Redshift database name	Type a name for the Redshift data warehouse.
Redshift cluster type	Choose a cluster type for the data warehouse. Default is single-node. For more information about Redshift cluster types, see Amazon EMR documentation.
Redshift cluster number of nodes	Number of compute nodes in the Redshift cluster. For a single node cluster type, the minimum is 1. For other cluster types, the minimum must be greater than 1.
Redshift cluster node type	Redshift cluster node instance type. Default is ds2.xlarge.
Redshift cluster user name	Type the username that is associated with the master user account for the Redshift cluster.
Redshift cluster password	Type a password for the Redshift cluster master user account.

After you finish entering values for the parameters, click **Next**.

AWS begins provisioning resources according to the values you entered.

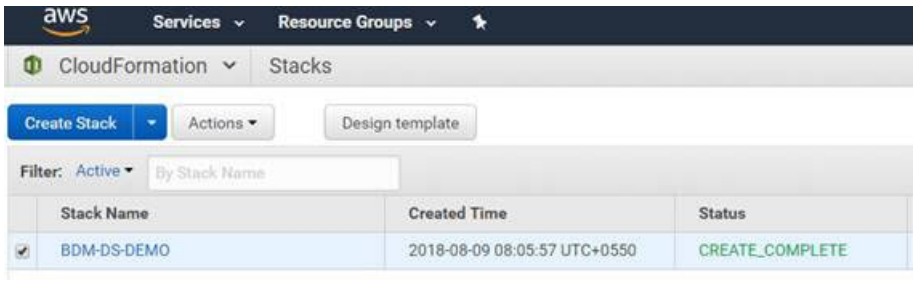
Monitoring Instance Provision and Informatica Domain Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain on the cloud platform.

AWS Resources

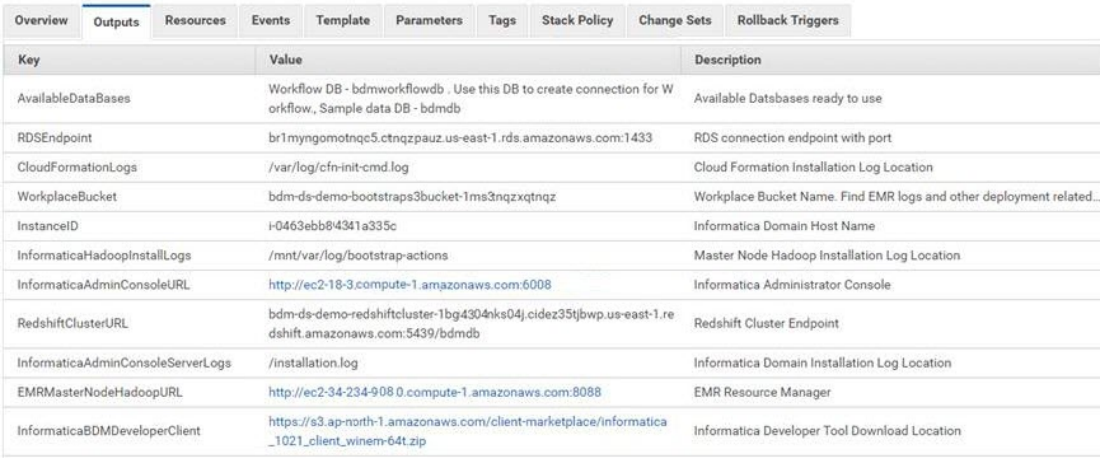
You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal.

The following image shows the display when the solution deployment is complete:



Click the **Output** tab to view the list of resources that the deployment process created in the VPC.

The following image shows the list of resources:



Informatica Resources

The list of deployed resources includes the **InformaticaAdminConsoleURL**. Click the linked value of this property.

The following image shows an example of this property:



The Administrator tool opens in a browser window.

Use the Informatica Administrator user name and password to log in to the Administrator tool.

You can use the Administrator tool to view the Informatica application services and resources that the automated deployment created.

Logs

Click the Logs tab in the Administrator tool to view the installation logs that the automated deployment created.

Installation log

Records the installation of the Informatica domain, Informatica services, and repositories. In case any of the installation tasks failed, you can check the log for the command that failed.

Filename: `Informatica_10.2.1_Services_<timestamp>.log`

Location: `/home/ec2-user/Informatica/server/`

Logs

After the completion of Big Data Management deployment, consult logs to see the success or failure of solution element creation.

You can access the following logs on the Informatica domain VM:

Command execution log

This log records the following events:

- Creation of Informatica connections, cluster configurations, and services.
- Population of the data warehouse and SQL databases.
- Import of sample mappings to the Model repository. This is recorded in the Project Importing section of the log.
- Data Integration Service recycling to register all changes.

At the top of the log file is a summary section that lists automated tasks and their status. Beneath the summary section are detailed sections about each task. If any of the tasks failed to complete successfully, you can look at the detailed section for the task to troubleshoot the task.

Filename: `Oneclicksolution_results.log`

Location: `/home/ec2-user/`

All of the following logs are located on the Informatica domain in `/home/ec2-user/individual_logs`.

Connection_creation.log

Lists the connection status of all connections that the automated deployment process tried to create.

DBcreation.log

Lists the following information about the RDS SQL Server database:

- MSSQL URL (connect string). For example:

  ```
  jdbc:informatica:sqlserver://bgee35ofgbm6dn.djffjnijpauz.us-east-1.rds.amazonaws.com:
  1433
  ```

- MSSQL user.
- MSSQL password.
- Domain DB. Name of the domain repository database.
- MRS DB. Name of the Model repository database.

InfaServiceLog.log

Describes the status of the cluster configuration, which the domain requires to enable communication between the domain and AWS EMR cluster resources.

CCO_Creation_Output.log

Records the output of cluster configuration and Hadoop connection creation commands.

MAS_creation.log

Records the creation and enablement of the MAS service.

WF_Setup.log

Records the execution of workflow connection and content creation commands.

Project_import.log

Lists the objects that the solution imported to the Model repository.

RDS_DB_load.log

Shows the status of execution of SQL commands to load the SQL database.

Redshift_DB_load.log

Shows the status of execution of SQL commands to load the data warehouse.

Post-Implementation Tasks

After the marketplace solution is finished deploying on AWS, perform the tasks in this section.

Using the Pre-Installed Mappings

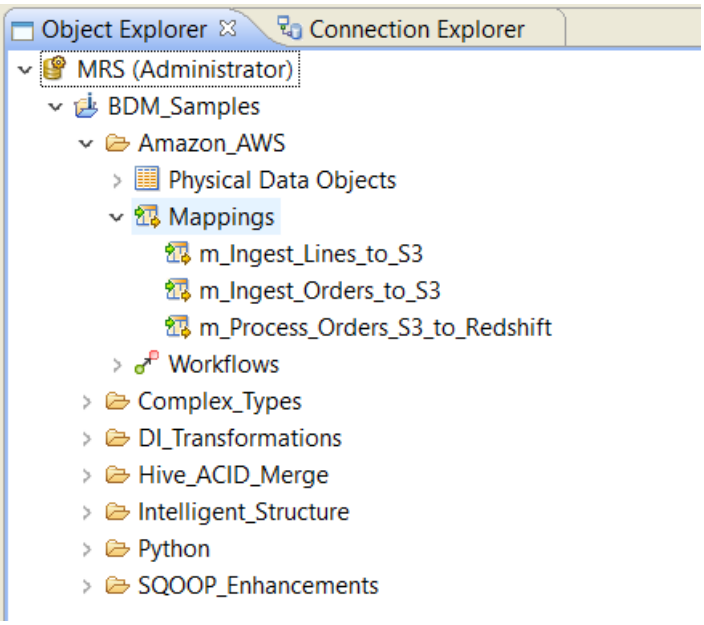
The marketplace solution contains sample pre-configured mappings that you can use as templates for your own mappings.

This section lists and describes the mappings and contains instructions for how to run them.

Pre-Installed Mappings

Use the Developer tool to open and run the pre-installed mappings that the automated deployment contains.

Browse the folders in the BDM_Sample project to access the pre-installed mappings. The following image shows the mappings inside the project folder:



The following table lists pre-installed sample mappings that you can run on the Amazon EMR cluster:

Folder	Mapping Name	Demonstrates
Amazon_AWS	m_Ingest_Lines_to_S3	Writing mapping results to S3
Amazon_AWS	m_Ingest_Orders_to_S3	Writing mapping results to S3
Amazon_AWS	m_Process_Orders_S3_to_Redshift	Reading from S3 and writing data to Redshift
Complex_Types	m_Create_Array_of_Structs	Creating nested complex types by creating an array of structs
Complex_Types	m_Create_Arrays	Creation of array data types
Complex_Types	m_Create_Structs	Creation of struct data types
Complex_Types	m_Process_Array	Using arrays
Complex_Types	m_R2H	Building hierarchical structure from relational sources
Complex_Types	m_R2H_Mini	Building simple hierarchical structures from relational sources
DI_Transformations	ALL_DI_Transformation	How to use all data integration transformations in Hadoop pushdown mode
Hive_ACID_Merge	Hive_Acid_Merge	How to use Update Strategy transformation on Hive tables that have transaction capability enabled on supported Hadoop distributions
Intelligent_Structure	m_ISD	How to use models exported from Intelligent Structure Discovery

Folder	Mapping Name	Demonstrates
Python	m_Python	Invoking data science model usage of Python transformation in a Big Data Integration mapping
Python	M_Titanic_PredictiveAnalysis_LoadHive	How to use Python based data science models to predict mortality rate of Titanic passengers
Sqoop_Enhancements	m_mssql_to_HDFS	Ingestion of data from Microsoft SQL Server (via SQOOP) into HDFS
Sqoop_Enhancements	m_mssql_to_Hive	Ingestion of data from Microsoft SQL Server (via SQOOP) into Hive

Ready To Go Scripts

Use a Ready To Go script to prepare the pre-installed mappings to run in your environment.

The automated deployment populates the `/home/ec2-user/ready_to_go` directory on the Informatica domain machine with the following files:

user.input

The user.input file contains a series of properties that you populate with information about cluster storage locations, including authentication information, which the pre-installed mappings require to run successfully.

ready_to_go.sh

The ready_to_go.sh script file edits the pre-installed mappings with values from the input file.

It is not necessary to edit the script file.

Run the Mappings

Browse to the mapping in the Developer tool and double-click the mapping to open it in the editor.

Each of the included mappings is configured to use the Spark engine. You must use the Spark engine to run the pre-configured mappings.

To run a mapping, select it in the list of mappings in the Navigator, and choose **Run** > **Run**.

The log file `/home/ec2-user/ReadyToGo.log` records the results of Ready to Go script execution.

Author

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